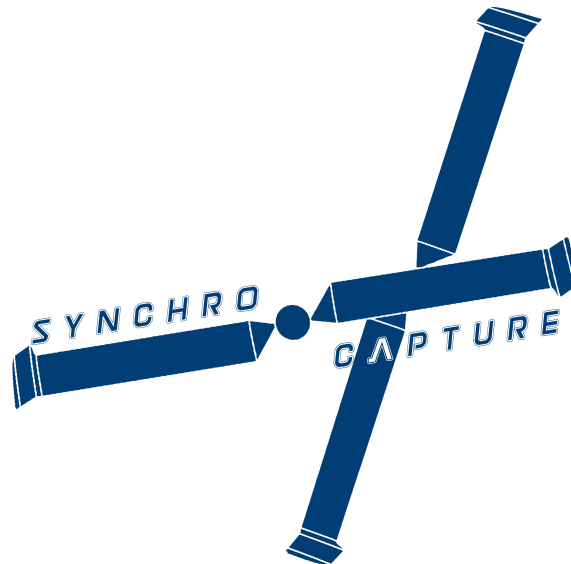


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

**Rotor Blade Aerodynamics and Aeroelasticity
Modelling for a Novel Synchropter Concept for
Aerial-Recovery of Falling Rocket Stages**



Student: Enson Tin-Hang Un
CID: 01355016

Project Supervisor: Dr. E. Levis
Team Supervisor: Dr. F. Montomoli
Submission Date: 15/06/2020

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ

Abstract

This report presents the aerodynamic and aeroelastic analysis of the rotor blade for a synchropter concept. The detailed methodology of implementing the combined blade element and momentum theory (BEM) is explained. My contribution to the project is summarised. According to the analysis, our helicopter is predicted to have a hover ceiling of 14,000ft on a regular day and 12,000ft on a hot day with full payload. It managed to achieve the maximum airspeed of 225km/h or 62.5m/s as promised. Data demonstrated a vertical climb rate of up to 1,968.5ft/min or 10m/s, superior to other commercially available models. The quality of the performance predictions is evaluated, with the discrepancies between teams discussed. Shortage of the current theory is reviewed. And potential areas for further development are spotted.

WRR Team Members

Table 1: Team Members and Roles

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

*Sub-team leads are indicated in **bold**.*

Contents

1	Introduction	1
2	Theory	1
2.1	Momentum Theory	1
2.2	Induced Velocity in Vertical Flight	2
2.3	Linear Inflow Approximation in Forward Flight	2
2.4	Combined Blade Element and Momentum Theory (BEM)	3
2.5	Aero-elastic Effects	4
2.6	Tip Loss Models	6
2.7	Intermeshing Rotor and Fuselage Losses	6
3	Results	7
3.1	BEM Code	7
3.2	Rotor Optimisation	7
3.3	Performance Prediction	9
3.4	Impacts	10
4	Discussion	10
5	Conclusion	12
	References	13
A	The BEM Code	15
B	Power Lapse Curves	19

List of Tables

1	Team Members and Roles	ii
2	Trimmed flight results at $V_x = 62.5\text{ms}^{-1}$, $V_z = 0.3\text{ms}^{-1}$	7
3	Final blade parameters	7
4	Rotor design specifications	7

List of Figures

1	Actuator disk model	1
2	Momentum theory results for the induced velocity in vertical flight [10]	2
3	Sectional view of a blade element (Modified from [10])	3
4	Power correction for tandem rotor in hover [12]	7
5	Trimmed flight results at $V_x = 62.5\text{ms}^{-1}$, $V_z = 0.3\text{ms}^{-1}$	8
6	Amplitudes of different harmonic modes of w at the tip in ψ direction	8
7	Hover ceiling vs maximum vehicle mass	9
8	Power requirement at different forward speeds at different altitudes	9
9	Power requirement at different vertical speeds at different altitudes	9
10	Schematics of the BEM code	12
11	Power lapse curve at $\Delta T = 0\text{K}$, 20K , 40K	19

Nomenclature

Physics Constants

ΔT Ambient temperature higher than that in ISA

ρ Ambient density of air

a Ambient speed of sound

g Acceleration due to gravity

Rotor Geometry and Parameters

$\dot{m}$ Mass flow rate through the rotor disk

η Rotor propulsive efficiency (defined in Eq.22)

κ_{ov} Induced power overlap correction factor

Ω Rotor angular velocity in rad/s, constant throughout flight

A Rotor disk area ($= \pi R^2$)

D Rotor diameter ($= 2R$)

d Separation distance between the centres of the two rotors

dm Blade mass per unit span

E Blade material Young's modulus

G Blade material shear modulus

I Blade moment of inertia about the root

J Blade torsional constant

N Number of blades in the rotor

R Rotor radius ($= D/2$)

r_0 Rotor hub radius

Blade Variables

α Local angle of attack

α_0 Zero-lift angle of attack

β Blade flapping deflection angle ($= \partial w / \partial r$)

Λ Sectional sweep angle

λ Inflow ratio ($= (V_z + v_i) / \Omega R$)

Λ_0	Sweep angle of the blade when stationary
λ_i	Induced inflow ratio ($= v_i/\Omega R$)
μ	Advance ratio ($= V_x/\Omega R$)
ϕ	Inflow angle
ψ	Blade azimuth position, zero at downstream direction
$\theta(r, \psi)$	Twist distribution ($= \theta_0 + \theta_c r \cos \psi + \theta_s r \sin \psi$)
θ_0	Collective pitch angle, positive when moveing leading edge upwards
θ_c, θ_s	Cyclic pitch angles, positive when moveing leading edge upwards
$c(r)$	Chord distribution
$C_D(\alpha, M, Re)$	Aerofoil sectional drag coefficient ($= 2dD/\rho V^2 c$)
C_{D0}	Zero-lift drag coefficient
$C_L(\alpha, M, Re)$	Aerofoil sectional lift coefficient ($= 2dL/\rho V^2 c$)
C_{Lmax}	Maximum lift coefficient
$C_M(\alpha, M, Re)$	Aerofoil sectional pitching moment coefficient ($= 2dM_r/\rho V^2 c^2$)
M	Local Mach number ($= V/a$)
M_{eff}	Local effective Mach number ($= M \cos \Lambda$).
r	Blade radial position, zero at the centre of the shaft
V	Local/sectional velocity magnitude
v_h	Induced velocity in hover, v_i in hover
v_i	Induced velocity at the rotor disk, positive when pointing downwards
V_x	Horizontal velocity experienced by the rotor, positive in forward flight
V_z	Vertical velocity experienced by the rotor, positive in climb
v_{flap}	Velocity caused by blade flapping motion ($= \dot{w}$), positive when pointing downwards
$w(r, \psi)$	Blade flapping displacement, normal to the rotor plane and positive when deflecting upwards
w_i	Induced velocity at far downstream, positive when pointing downwards

Forces, Moments and Power

B	Effective span factor/Tip loss factor
BM	Beam bending moment

C_T	Horizontal velocity experienced by the rotor, positive in forward flight
dF_r	Sectional force in the radial direction, positive with pointing away from the centre
dF_x	Sectional force in the rotor disk plane, positive when acting opposite to the rotor direction of motion
dF_z	Sectional force normal to the rotor disk plane, positive when acting upwards
dM_r	Sectional twisting moment, positive when producing positive twist
$F(r)$	Tip loss function
H	Horizontal force, in the direction of horizontal velocity and positive when directed rearwards
M_x	Rolling moment, positive when the retreating side moves downwards
M_y	Pitching moment, positive when pitching up
P	Shaft power, positive when power is supplied to the rotor
$P_{isolated}$	Shaft power when the rotor is separated infinitely apart
P_{total}	Total power supplied by the engine
SF	Beam shear force
T	Thrust, acting normal to the rotor plane and positive when pointing upwards
Y	Side force, acting in the rotor plane perpendicular to the horizontal velocity direction and positive when pointing to the advancing side

Abbreviation

DL	Disk loading ($= T/A$)
FM	Figure of merit
GTOW	Gross takeoff weight ($= \text{MTOW} - \text{weight of payload}$, 1507kg)
ISA	International standard atmosphere model
MTOW	Maximum takeoff weight (2907kg)
VRS	Vortex Ring State

1 Introduction

This project aims to develop an unmanned synchropter for aerial-recovery of falling rocket stages. The responsibilities of the Aerodynamics Team are 1) Sizing of the rotor, 2) Aerofoil selection and optimisation, and 3) Vehicle aerodynamic performance prediction with and without payload [3]. My role in the team is about developing and testing the computational method to reliably simulate the rotor performance.

2 Theory

2.1 Momentum Theory

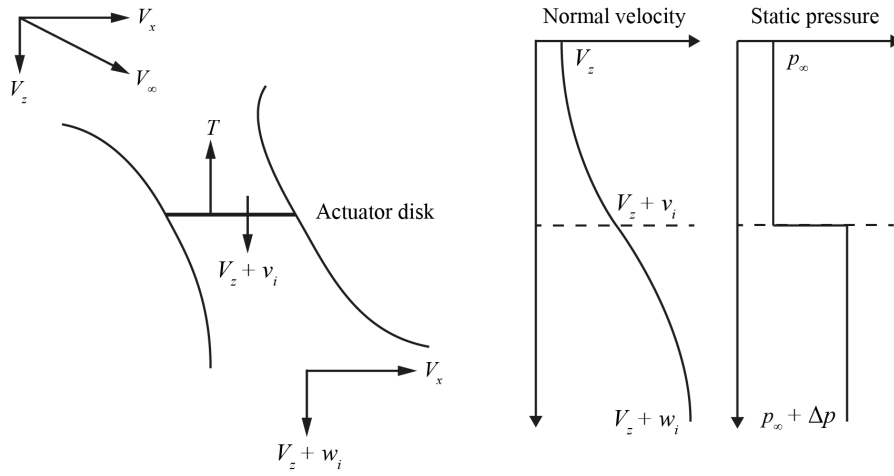


Figure 1: Actuator disk model

The analysis begins by considering one rotor in a freestream V_x and V_z as illustrated by Fig.1. Momentum theory idealises a rotor as an infinitesimally thin actuator disk which exerts a force $-T$, normal to the disk, on the air. Velocity is continuous but a static pressure jump occurs across the disk. Conservation of mass, momentum and energy give

$$\dot{m} = \rho A \sqrt{V_x^2 + (V_z + v_i)^2}, \quad (1)$$

$$T = \dot{m} w_i, \quad (2)$$

and

$$P = T(V_z + v_i) = \frac{1}{2} \dot{m} [V_x^2 + (V_z + w_i)^2] - \frac{1}{2} \dot{m} [V_x^2 + V_z^2]. \quad (3)$$

respectively. All symbols have their meaning as defined in the nomenclature.

Combining Eq.1-3 yields $w_i = 2v_i$, and

$$v_i = \frac{T}{2\rho A \sqrt{V_x^2 + (V_z + v_i)^2}}. \quad (4)$$

Therefore, we concluded that there must be some induced velocity if thrust is generated.

Notice that our analysis assumes the flow velocity through the rotor $V_z + v_i$ to be pointing downwards. In reality, $V_z + v_i$ (and hence the shaft power) changes sign in descent when the rotor passes the $V + v_i = 0$ line (v_i is labelled v in Fig.2). As the rotor passes the $V + 2v_i = 0$ line, the far wake velocity changes sign, i.e. the rotor becomes a wind turbine. Fig.2 confirms that our theory is valid for climb and small descent rates before the rotor enters an unstable region called vortex ring state. We will avoid going into VRS, so, the theory fits our purpose.

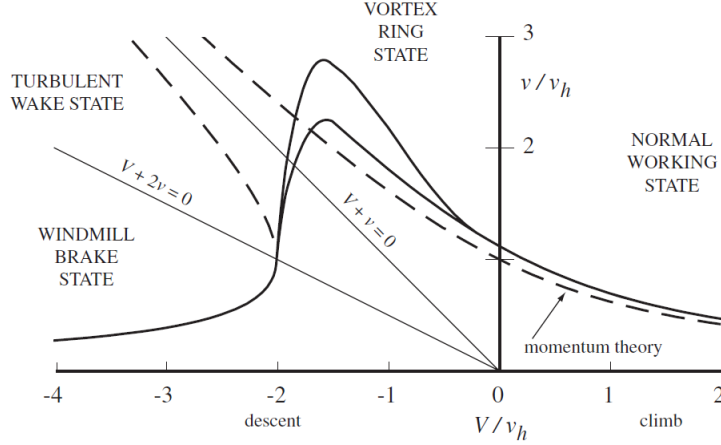


Figure 2: Momentum theory results for the induced velocity in vertical flight [10]

2.2 Induced Velocity in Vertical Flight

Now consider the case of vertical flight, i.e. $V_x = 0$, Eq.4 is simplified into

$$v_i = -\frac{V_z}{2} + \sqrt{\left(\frac{V_z}{2}\right)^2 + \frac{T}{2\rho A}}. \quad (5)$$

This result is not adequate as it only provides an average induced velocity. We extend the equation to a non-uniform inflow by considering its differential form. Due to the symmetry around the shaft axis, the thrust produced in an infinitesimal annulus area $dA = 2\pi r dr$ is a function of radial position r only, so does v_i .

$$v_i(r) = -\frac{V_z}{2} + \sqrt{\left(\frac{V_z}{2}\right)^2 + \frac{dT}{2\rho dA}}. \quad (6)$$

2.3 Linear Inflow Approximation in Forward Flight

For forward flight, converting Eq.4 into a non-uniform inflow distribution is not as straightforward as the previous section. This is because the inflow is no longer axial-symmetric, and now lift does not only depend on r but also the azimuth angle ψ . Many studies [10, 16, 18] suggest that a linear inflow model can decently approximate the real inflow structure, i.e.

$$v_i(r, \psi) = \bar{v}_i(1 + \kappa_x r \cos \psi + \kappa_y r \sin \psi), \quad (7)$$

where $\bar{v}_i$ is the mean inflow velocity calculated from Eq.4.

The most commonly used model is given by Coleman, Feingold, and Stempin [5],

$$\kappa_x = \sqrt{1 + \left(\frac{\lambda}{\mu}\right)^2} - \left|\frac{\lambda}{\mu}\right|. \quad (8)$$

This expression, however, has an inherent singularity when μ approaches zero, i.e. when transforming between vertical and forward flight. In order to seek a complete model which remains valid in all flight conditions, another approximation by Drees [6] is applied:

$$\kappa_x = \tan \frac{\chi}{2} = \tan \left(\frac{1}{2} \tan^{-1} \frac{\mu}{\lambda} \right), \quad (9)$$

$$\kappa_y = -2\mu. \quad (10)$$

In practice, numerical instability is encountered at small values of μ and λ . Hence, a relaxation factor $0 < \kappa_{rx} \leq 1$ is introduced in updating v_i , such that $(v_i)_{new} = \kappa_{rx}(v_i)_{old} + (1 - \kappa_{rx})(v_i)_{new}$.

2.4 Combined Blade Element and Momentum Theory (BEM)

In contrast to the momentum theory, which treats the entire rotor as a homogeneous disk, the blade element theory takes into account the blade geometry, such as chord and twist.

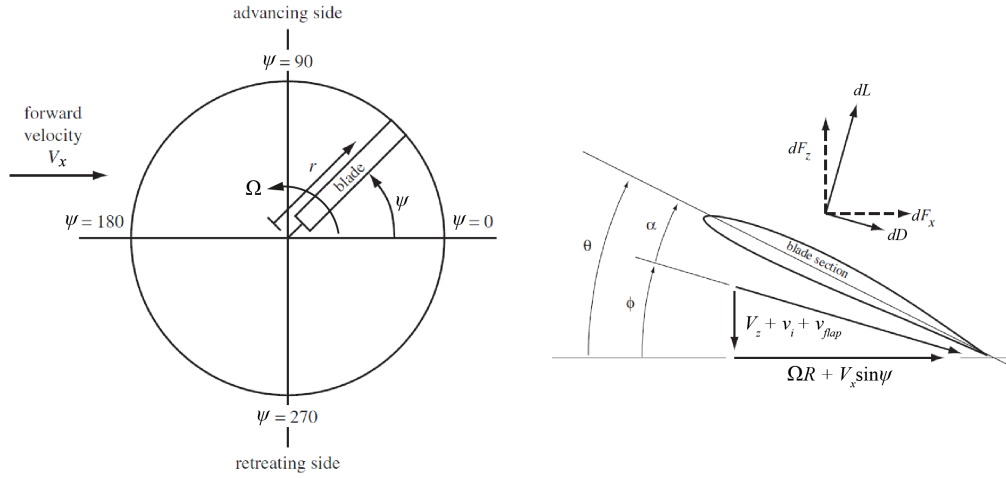


Figure 3: Sectional view of a blade element (Modified from [10])

In blade element theory, a blade is discretised into many small ‘blade elements’, each acts as a 2D wing section. Sectional lift and drag are calculated using C_L and C_D of the chosen aerofoil section. C_L and C_D data are obtained from CFD simulations performed by Kevin Yin [20]. The local angle of attack α and velocity V are calculated by

$$\alpha(r, \psi) = \tan^{-1} \left(\frac{V_z + v_i + v_{flap}}{\Omega r + V_x \sin \psi} \right) - \theta, \quad (11)$$

$$V(r, \psi) = \sqrt{(V_z + v_i + v_{flap})^2 + (\Omega r + V_x \sin \psi)^2 + (V_x \cos \psi)^2}. \quad (12)$$

However, the blade element theory provides no information about the induced velocity v_i . Therefore, it is calculated using the momentum theory result from the preceding sections. This gives

rise to the name of combined blade element and momentum theory. The total thrust T and shaft power P are computed by numerical integration over the span, then averaged over $\psi \in [0, 2\pi]$:

$$dF_z = \frac{1}{2} \rho c V^2 (C_L \cos \phi - C_D \sin \phi), \quad (13)$$

$$dF_x = \frac{1}{2} \rho c V^2 (C_L \sin \phi + C_D \cos \phi), \quad (14)$$

$$dF_r = -dF_z \sin \beta, \quad (15)$$

$$T = \frac{N}{2\pi} \int_0^{2\pi} \int_{r_0}^R dF_z \cos \beta \, dr \, d\psi, \quad (16) \quad P = \frac{N}{2\pi} \int_0^{2\pi} \int_{r_0}^R dF_x \Omega r \, dr \, d\psi, \quad (19)$$

$$H = \frac{N}{2\pi} \int_0^{2\pi} \int_{r_0}^R dF_r \cos \psi + dF_x \sin \psi \, dr \, d\psi, \quad (17) \quad M_x = \frac{N}{2\pi} \int_0^{2\pi} \int_{r_0}^R dF_z r \sin \psi \, dr \, d\psi, \quad (20)$$

$$Y = \frac{N}{2\pi} \int_0^{2\pi} \int_{r_0}^R dF_r \sin \psi - dF_x \cos \psi \, dr \, d\psi, \quad (18) \quad M_y = -\frac{N}{2\pi} \int_0^{2\pi} \int_{r_0}^R dF_z r \cos \psi \, dr \, d\psi. \quad (21)$$

The effect of sweep is accounted by a reduced effective Mach number $M_{eff} = M \cos \Lambda$, where sweep angle $\Lambda = \Lambda_0 + \tan^{-1}(V_x \cos \psi / (\Omega r + V_x \sin \psi))$.

The propulsive efficiency η is defined as

$$\eta = \frac{P_{out}}{P_{in}} = \frac{T(V_z + v_i)}{P}. \quad (22)$$

In the special case of hover, where $V_z = 0$, η is named figure of merit (abv. FM). $FM = C_T^{1.5} / \sqrt{2} C_P$. A FM of 0.7 is used for initial power sizing in the early design phase.

2.5 Aero-elastic Effects

In forward flight, an additional velocity is seen by the blade elements. This is caused by the blade deformation. A blade deforms by flapping (out of plane), lagging (in-plane), radial elongation and rotation around its span axis. However, the out of plane and radial forces are observed to be at least one order of magnitude greater than that in-plane, and the radial deformation has minimal effect, if any, on the aerodynamic forces. Thus, only the flapping and twisting motions are studied. Throughout this section, the blade is modelled as a Euler-Bernoulli beam, which means the deflections $\Delta\theta$ and w are small, and the neutral axis has a constant position and remains perpendicular to the cross-sections.

The twisting motion is caused by the pitching moment $dM_r(r)$ of the aerofoil section. If the flexural axis, the line joining all sectional shear centres, is align with the centre of mass and the aerodynamic centre of each section, the pitching moment is only a function of the local velocity, i.e. $dC_M/d\alpha = 0$. The twisting deformation can be expressed by

$$\Delta\theta(r) = \int_{r_0}^R \frac{\rho c^2 V^2 C_M}{2GJ} \, dr. \quad (23)$$

Two forces are involved in blade flapping, the normal force dF_z and the sectional self-weight dmg . Its dynamical behaviour is formulated by the Euler–Lagrange equation [13], that is

$$EI \frac{\partial^4 w}{\partial r^4} + dm \frac{\partial^2 w}{\partial t^2} = dF_z - dm g. \quad (24)$$

The equation is solved for vertical deflection $w(r, \psi)$ using a multi-step method as follows.

$$SF(r, \psi) = EI \frac{\partial^3 w}{\partial r^3} = \int_{r_0}^R dF_z - dm \left(\frac{\partial^2 w}{\partial t^2} + g \right) dr, \quad (25a)$$

$$BM(r, \psi) = EI \frac{\partial^2 w}{\partial r^2} = \int_{r_0}^R SF dr, \quad (25b)$$

$$\beta(r, \psi) = \frac{\partial w}{\partial r} = \int_{r_0}^R \frac{BM}{EI} dr, \quad (25c)$$

$$w(r, \psi) = \int_{r_0}^R \beta dr. \quad (25d)$$

The vertical velocity caused by blade flapping is then calculated using w , as

$$v_{flap} = \frac{\partial w}{\partial t} = \frac{\partial w}{\partial \psi} \cdot \frac{\partial \psi}{\partial t} = \Omega \frac{\partial w}{\partial \psi}. \quad (26)$$

This flapping velocity typically has the same order of magnitude as the induced velocity v_i . Consequently, it has considerable effect in α , and hence, dF_z . Hence, a circular dependency between dF_z and w is established, which requires w to be C^∞ in the ψ direction to prevent singularity from emerging as the number of iterations grows large.

Attempts of applying a 1D convolution kernel over w in the ψ direction were made. E.g. if the kernel used is $[0.4, 0.2, 0.4]$, then

$$w(r, \psi_k) = 0.4w(r, \psi_{k-1}) + 0.2w(r, \psi_k) + 0.4w(r, \psi_{k+1}). \quad (27)$$

This method was not successful as singularities still appear eventually.

At this point, the sudden recollection of a classic class of smooth functions, $\sin n\psi$ and $\cos n\psi$ $\forall n \in \mathbb{N}$, has shifted the attention to truncated Fourier series, i.e.

$$w(r, \psi) \approx A_0(r) + \sum_{k=1}^K A_k(r) \cos\left(\frac{k\psi}{2}\right) + B_k(r) \sin\left(\frac{k\psi}{2}\right). \quad (28)$$

With M data points in the ψ direction, the Fourier coefficients are approximated by

$$A_k(r) \approx \frac{2}{M} \sum_{m=1}^M w(r, \psi_m) \cos((k-1)\psi_m), \quad (29)$$

$$B_k(r) \approx \frac{2}{M} \sum_{m=1}^M w(r, \psi_m) \sin((k-1)\psi_m). \quad (30)$$

Since w is naturally periodic in the ψ direction, in theory, the truncated Fourier series will converge to the exact solution for sufficiently large K . However, as K increases, extensive oscillations were observed similar to the Gibbs phenomenon [7]. This is suspected to be caused by using summations to approximate the integrals in determining the Fourier coefficients. The solution to this problem is to search for a large K , then only include the dominating harmonic modes (those with the largest Fourier coefficients), and set the other modes to zero.

2.6 Tip Loss Models

3D flow effects caused by the finite span of the blades are neglected in BEM. In particular, we concern about the problem that the tip is producing non-zero lift, which violates the lifting line theory as $d\Gamma/dr = \infty$ at the tip. A rigorous solution can be obtained by analytical or numerical analysis of the wake, e.g. a free-vortex wake model [17], but this is deemed beyond the scope of this project. So, instead, two approximation methods are considered.

The first one is the effective span factor B proposed by Prandtl [15]. For a low inflow rotor,

$$B = 1 - 2 \ln 2 \frac{\sqrt{C_T}}{N}. \quad (31)$$

Any spanwise position $r/R > B$ is assumed to produce no lift but drag.

The second approximation was suggested by Betz, Prandtl and Glauert [8, 15] based on asymptotic analysis of a vortex sheet wake model. Lift is multiplied by a function $F(r)$ which has the value of one at the root and drops to zero as it approaches the tip.

$$F(r) = \frac{2}{\pi} \cos^{-1} \left(\exp \frac{(r-1)N}{2\lambda_i} \right). \quad (32)$$

Since F is a function of v_i and v_i is dependent on the lift, a solution is reached by iteration. Despite being more computationally expensive, the second model is adopted as it gives more detailed lift distribution, which is essential for calculating the flapping deflection w and is beneficial for structural analysis.

2.7 Intermeshing Rotor and Fuselage Losses

In our synchropter, the two counter-rotating rotors are separated by a distance of $d = 1.52\text{m}$ and are tilted by 14° away from each other. The flow induced by the advancing side of one rotor is blown on the retreating side of the other. Since the rotors are tilted, the wake of the upper rotor hits the lower one at an angle. The wake will also be redirected by the lower rotor and will interfere with the wakes from both rotors and with the fuselage. Analysis of such systems is rarely seen in the literature. To simplify this complicated problem, we decided to analysis the effect of tilting, overlapping and fuselage drag separately.

The effect of tilting is reflected by multiplying the total lift force, that counteracts the weight in trim flight, by a factor of $\cos 14^\circ$.

Overlapping of two rotors results in an additional induced power. The total power $P_{total} = \kappa_{ov} P_{isolated}$, where κ_{ov} is given by Leishman [12] in Fig.4 and by Johnson [10] in Eq.33.

$$\kappa_{ov} = \frac{2}{(1 + d/D)^2}. \quad (33)$$

Finally, the drag induced by the fuselage in the wake is calculated using the drag coefficient calculated by Alessandro Polvara [14], ambient density ρ and the velocity through the rotor disk $V_z + v_i$. At hover, this is about 5.2% of the thrust produced by the rotor. The fuselage induced drag is less important in forward flight as v_i gets smaller, but it is very pronounced in climb.

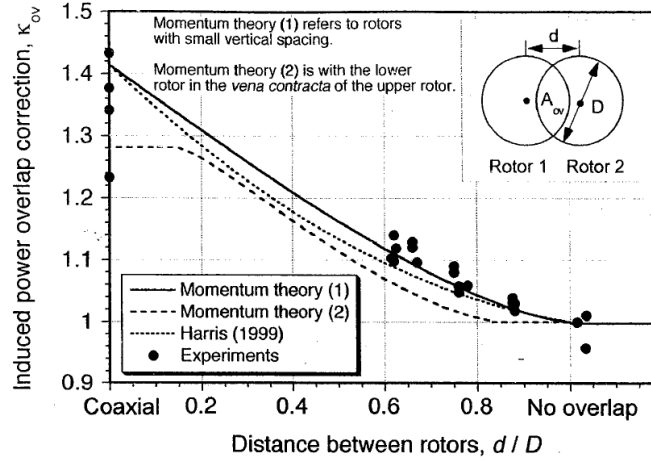


Figure 4: Power correction for tandem rotor in hover [12]

3 Results

3.1 BEM Code

The major outcome of this project is the BEM code attached in Appendix A. Results of an example case (for one rotor, assuming no fuselage drag) is presented in Table 2 and Fig.5.

Table 2: Trimmed flight results at $V_x = 62.5\text{ms}^{-1}$, $V_z = 0.3\text{ms}^{-1}$

T [N]	H [N]	Y [N]	P [kW]	M_x [Nm]	M_y [Nm]	η
14535	74.06	-126.9	147.9	-6.957	-59.68	0.373

3.2 Rotor Optimisation

The blade optimisation was carried out collaboratively with Kelvin Cheng [4]. The optimisation problem aims to maximise propulsive efficiency in hover and at the maximum forward speed at 20000ft, subjected to the constraints that 1) no part of the blade, except that in the reverse flow zone, is stalled, and 2) lift must be greater than the helicopter's weight.

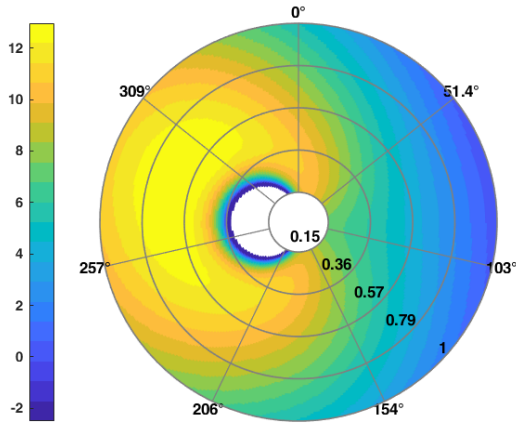
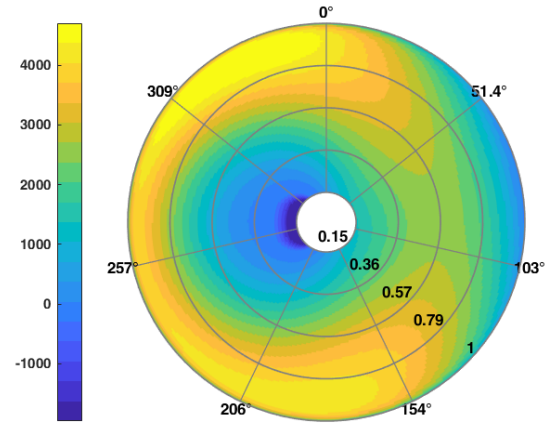
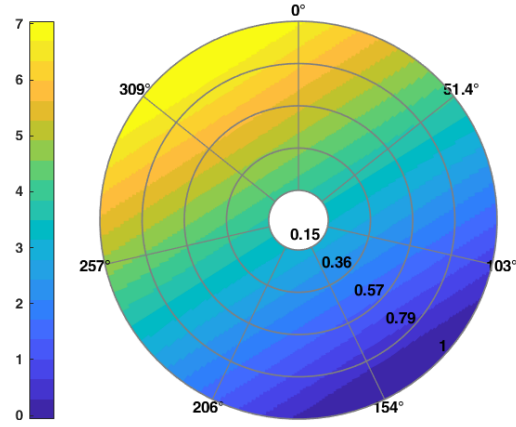
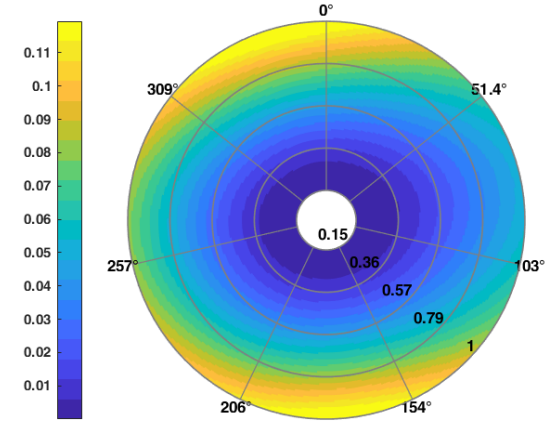
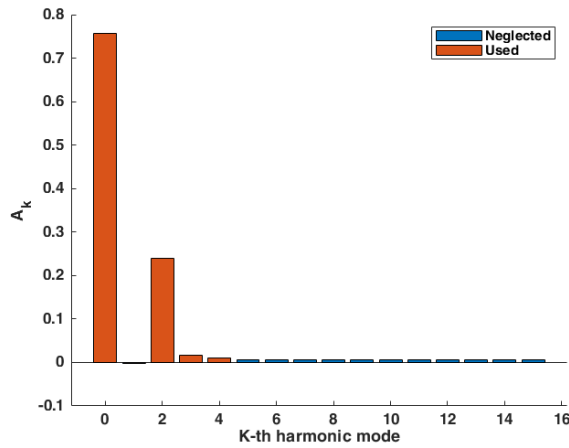
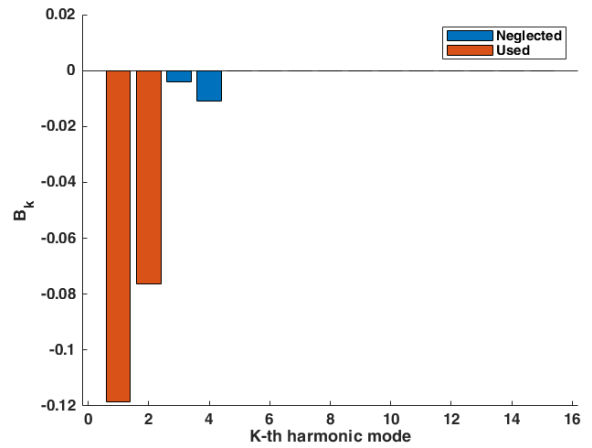
A numerical approach was implemented in MATLAB. We departed from a set of initial guesses, then converged to a few local optimum points. The global optimiser is approximated by the best local optimiser. The final result is shown in Table 3 and 4.

Table 3: Final blade parameters

Rotor radius	Root chord	Rotational speed Ω	Maximum tip Mach number
4.023m	57.4cm	206.05m/s	0.85

Table 4: Rotor design specifications

	Aerofoil	Taper ratio	Twist
Root section ($0 \leq r/R < 0.9$)	NACA23012	1	-11°
Tip section ($0.9 \leq r/R \leq 1$)	NACA63206	Varying (see [20])	-17° (with respect to root)

(a) Angle of attack α [deg](b) dF_z distribution [Nm^{-1}](c) Inflow velocity v_i [ms^{-1}](d) Blade deflection w [m]**Figure 5:** Trimmed flight results at $V_x = 62.5\text{ms}^{-1}$, $V_z = 0.3\text{ms}^{-1}$ (a) Cosine modes, $A_k \cos(k\psi/2)$ (b) Sine modes, $B_k \sin(k\psi/2)$ **Figure 6:** Amplitudes of different harmonic modes of w at the tip in ψ direction

3.3 Performance Prediction

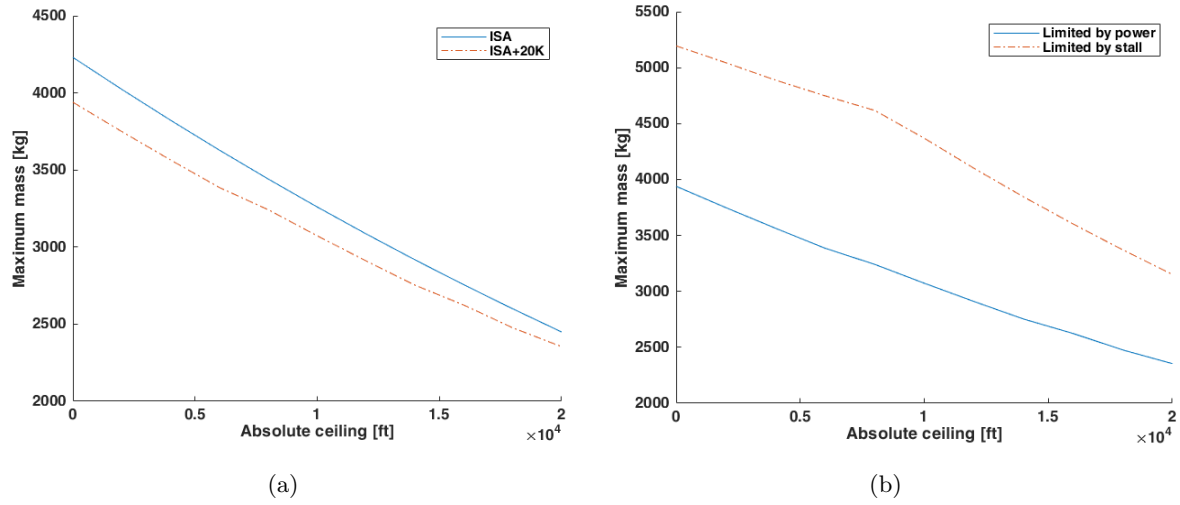
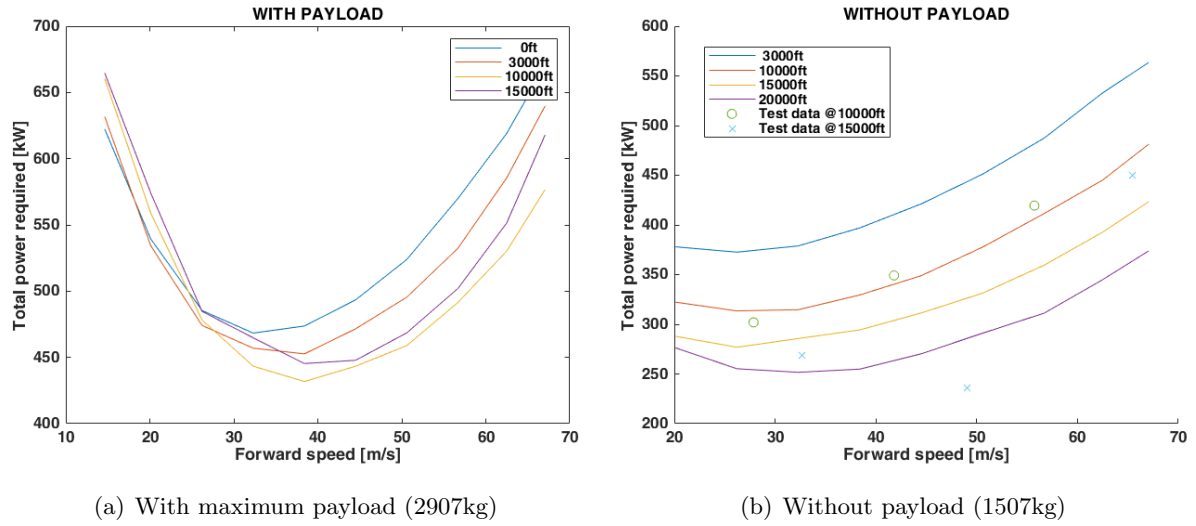


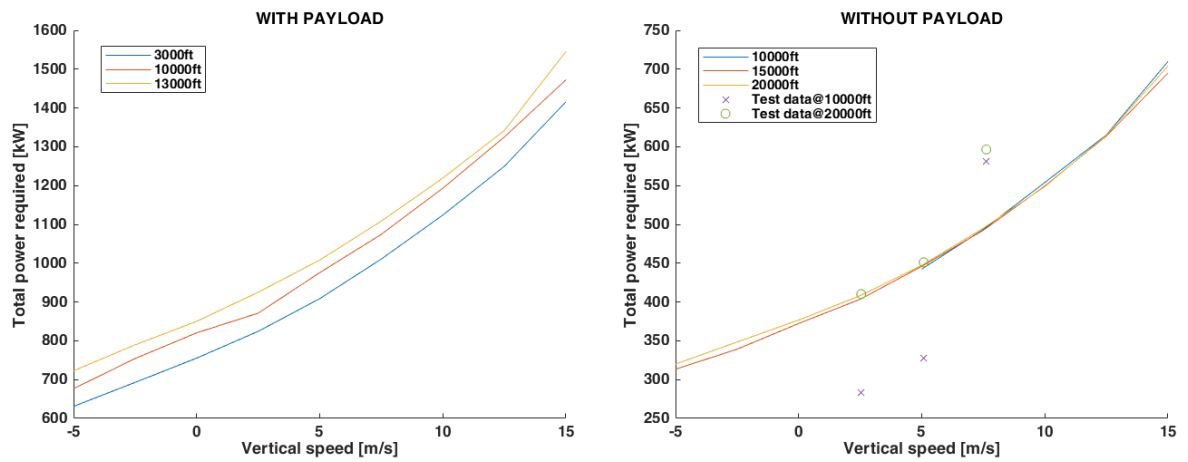
Figure 7: Hover ceiling vs maximum vehicle mass



(a) With maximum payload (2907kg)

(b) Without payload (1507kg)

Figure 8: Power requirement at different forward speeds at different altitudes



(a) With maximum payload (2907kg)

(b) Without payload (1507kg)

Figure 9: Power requirement at different vertical speeds at different altitudes

3.4 Impacts

The Aerodynamics Team works mainly on the upstream side of the aircraft design process. Our data have huge impacts on other design aspects, e.g.

1. The momentum theory analysis was used for initial rotor sizing,
2. The momentum theory analysis was shared with the Systems and Propulsion Team (hereinafter as Systems Team) for initial powerplant sizing,
3. Theory and implementation of BEM were shared with the Flight Mechanics, Performance and Controls Team (hereinafter as Flight Mech. Team) to aid their code development,
4. The BEM code was used in rotor optimisation,
5. Inflow data from the BEM code were shared within the team and with the Flight Mech. Team for analysing the horizontal stabiliser behaviour in the rotor wake,
6. The normal force distribution from the BEM code was shared with the Structures and CAD Team for structural analysis of the blade,
7. Twisting moment data were shared with the Systems Team for actuator sizing,
8. Forward flight data from the BEM code were shared with the Flight Mech. Team for range and endurance predictions, and determining the best cruise speed [9],
9. Climb performance data from the BEM code were shared with the Flight Mech. Team for analysing climb performance, and
10. Theoretical advice [2] was given to the Flight Mech. Team on predicting the region of VRS.

4 Discussion

What is confirmed?

Since the BEM code is the foundation of all performance and flight dynamics analysis, for safety purposes, it was developed independently with Kelvin Cheng [4]. The code was verified against test cases in Prouty [16], and the key results are consistent (within 10% difference) with Kelvin's model. Having a closer look into the example results in Section 3.1, it is observed that the BEM code has successfully resembled the key features of a helicopter rotor in forward flight: e.g. 1) The reverse flow zone of diameter $\approx \mu$ at the retreating root, 2) Higher angle of attack due to slower horizontal velocity on the retreating side and the opposite on the advancing side, and 3) inflow $v_i \approx 0$ in the front of the rotor disk when $\mu \gg \lambda$ [10].

Power predictions for forward flight and vertical flight are plotted Fig.8 and 9. They are compared with the graphs in literature [10, 12, 16, 18] and are found to have similar structures. X-Plane simulations are ran by Gordon Wong [19] to confirm the result. The test data are plotted in Fig.8(b) and 9(b). All test data agree with the predicted trend.

What is discrepant?

The second test data for forward flight power at 15000ft is significantly lower than the curve. It is believed to be caused by pilot errors as the other two data points at the same altitude and

those at 10000ft give good reconciliation of the prediction.

Moreover, although the data for vertical climb power match the predicted trend, the range of error is huge (about 50%). This can be explained by a feedback reported by the Simulation Team that maintaining purely vertical flight is extremely hard as the helicopter appears to yaw extensively at small forward speed. However, we see no theoretical explanation for such phenomenon. Even if the helicopter is in turbulence, the roll-pitch-yaw coupling would mean that there should not only be yaw motion but also roll and pitch motions. Therefore, we speculate that it may be the problem in the controller or the flight simulation software that caused this strange behaviour.

Furthermore, the hover ceiling with full payload is predicted to be 14,000ft on a normal day and 12,000ft on a hot day ($\Delta T = 20K$) using Fig.7. This result coincides with the calculation done by Omar Arangath [1] from the Systems Team. Yet, the hover ceiling found from simulations is 7,000ft – a tremendous mismatch with our result! A number of reasons were investigated. We first checked the method X-Plane used to model the fuselage. It turns out that the fuselage was handled as a point that will only experience drag in the forward flight, i.e. the drag coefficient is treated as a tensor with only one non-zero entry. This is different from our model, but neglecting the fuselage drag should have resulted in an even better ceiling performance. Next, we examined the C_L and C_D curves of the aerofoil sections. We discovered that the Simulation Team mistakenly entered the zero-lift angle of attack as 1° instead of -1° , and the tip aerofoil is not changed. The lift curve slope is the same, but C_{D0} is about 50% lower in their model. The difference in α_0 can easily be compensated by reducing θ_0 . And the lower C_{D0} together with the use of the root section, which has a higher lift curve slope and C_{Lmax} , are supposed to give better performance, not worse. It is confirmed by the Simulation Team that X-Plane does not model wave drag, ergo, the simulation should not be disadvantaged even if the higher M_{DD} and sweep of the tip section were not modelled. Finally, the attention was turned to the engine power. The power lapse model was checked to be correct, but the engine power output when hovering at 7000ft is reported to be only half of that available. The collective pitch is checked and the possibility of rotor stall is eliminated. To conclude, the exact reason for the discrepancy in the hover ceiling is still unclear by the submission of this report. It will be of the highest priority to resolve this problem in the future design iteration.

What is missing?

The major blind spot of the current theory is the BERP tip design. Further research can be done, e.g. how swept blade tip affects lift and drag distribution, how the stable vortex structure [11] around the leading edge affect the stall angle and possibly the post-stall behaviour, or how the misalignment of centre of gravity and aerodynamic centre varies the aeroelastic responses.

In addition, the rotor wake and its interaction with the rotor and the fuselage are neglected. Analysis can be extended by including a prescribed vortex wake model with the vorticities computed in a similar manner as the lifting line theory and the induced velocity determined using Biot–Savart law. Or even more advanced methods.

Furthermore, the effects of the intermeshing rotor are overly simplified by empirical correction

factors. Not to mention that lift is still calculated as the two rotors separated infinitely apart. These effects are essential to building a realistic model. However, as mentioned before, literature and industrial data about synchropters are rare.

What to improve in next design iteration?

Apart from fixing the hover ceiling problem, the most important task is to re-optimize the rotor blade. This is because the aircraft was initially required to hover with full payload at 20,000ft¹. With this in mind, the blade was optimized for this most constraining operating condition. Meanwhile, the Coordination Team and Systems Team decided that our engine only needs to be capable of supporting hover up to 10,000ft. As a result, as shown in Fig.7(b), the hovering performance is always limited by power, not stall. If we finally decided to change the mission profile and lower the ceiling requirement, the chord of the blade can be reduced, facilitating a higher R , which can lead to higher propulsive efficiency. Of course, a shorter chord will likely pose challenge on the Structures Team. Design iterations is therefore vital.

5 Conclusion

In conclusion, the BEM code is the primary result of this project. The theory and implementation behind it is reviewed in this report. My personal contribution to the project is outlined. Using the BEM code, the fully loaded helicopter is predicted to have a hover ceiling of 14,000ft on a normal day and 12,000ft on a hot day. It can achieve the design maximum speed of 225km/h or 62.5m/s at up to 15,000ft. The maximum vertical climb rate at 3,000ft (where the final attempt will be conducted) is 1,968.5ft/min or 10m/s, about three times faster than other helicopters in the class. The quality of the performance predictions are assessed and discussed. Potential areas for future works are identified.

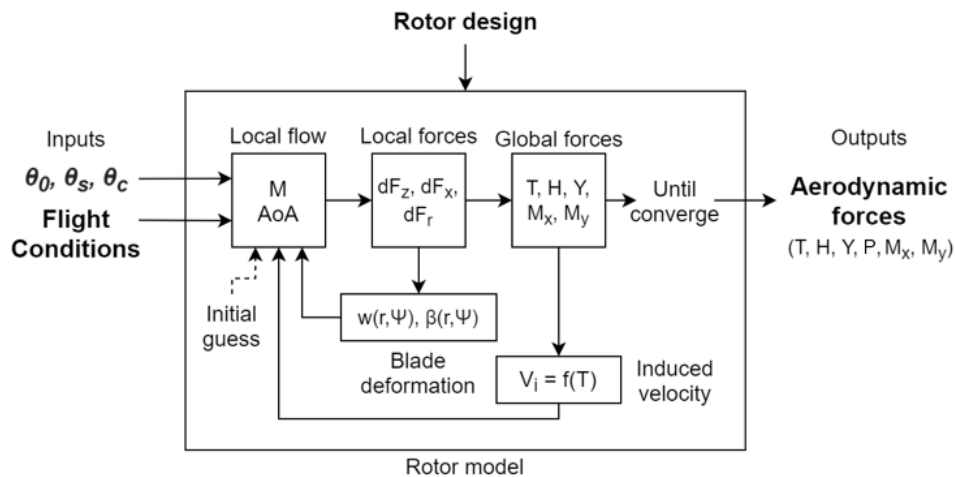


Figure 10: Schematics of the BEM code

¹This requirement is still in the mission requirement document by the time of submission of this report. I quote, 'recovery manoeuvre to be initiated at 20,000 ft.'

References

- [1] ARANGATH, O. Engine design for a novel synchropter for the aerial capture of an orbital launch system. GDP individual report, 2020.
- [2] BASSET, P.-M., CHEN, C., PRASAD, J., AND KOLB, S. Prediction of vortex ring state boundary of a helicopter in descending flight by simulation. *Journal of the American Helicopter Society* 53 (04 2008), 139–151.
- [3] BRUCE, P., AND LEVIS, E. *3rd-Year Group Design Project, Project Statement*. Department of Aeronautics, Imperial College London, 2020.
- [4] CHENG, K. C.-W. Rotor design of a novel synchropter concept for the aerial-recovery of falling rocket stages. GDP individual report, 2020.
- [5] COLEMAN, R. P., FEINGOLD, A. M., AND STEMPIN, C. W. *Evaluation of the Induced-Velocity Field of an Idealized Helicopter Rotor*. NASA, Langley Research Center, Hampton, VA., 1945.
- [6] DREES, J. M. A theory of airflow through rotors and its application to some helicopter problems. *Journal of the Helicopter Association of Great Britain* 3, 2 (1949), 79–104.
- [7] GIBBS, J. W. Fourier’s series. *Nature* 59 (1539), 606.
- [8] GLAUERT, H. *The elements of airfoil and airscrew theory*. Cambridge University Press, New York, NY., 1926.
- [9] HON, J. K.-S. Flight performance analysis of the novel synchropter. GDP individual report, 2020.
- [10] JOHNSON, W. *Rotorcraft Aeromechanics*. Cambridge University Press, New York, NY., 2013.
- [11] LEISHMAN, J. G. ENAE 632 - The British Experimental Rotor Program (BERP) Blade.
- [12] LEISHMAN, J. G. *Principles of Helicopter Aerodynamics (2nd Edition)*. Cambridge University Press, New York, NY., 2008.
- [13] PINHO, S. T. *Structural Dynamics*. Department of Aeronautics, Imperial College London, London., 2018.
- [14] POLVARA, A. Fuselage aerodynamics and slung load modelling of a novel synchropter concept for the aerial-recovery of falling rocket stages. GDP individual report, 2020.
- [15] PRANDTL, AND BETZ. *Vier Abhandlungen zur Hydrodynamik und Aerodynamik (German) [Four papers on hydrodynamics and aerodynamics]*. Göttinger Nachr, Göttinger, 1927.
- [16] PROUTY, R. W. *Helicopter Performance, Stability, and Control*. Krieger Publishing Company, Malabar, FL, 1986.

- [17] RIBERA, M. *Helicopter Flight Dynamics Simulation With a Time-Accurate Free-Vortex Wake Model*. PhD thesis, University of Maryland, College Park, Maryland, 2007.
- [18] STEPNIEWSKI, W. Z. *Rotary-Wing Aerodynamics - Volume I: Basic Theories of Rotor Aerodynamics (With Application to Helicopters)*. Boeing Vertol Company (Now Boeing Rotorcraft Systems), Philadelphia, PA, 1979.
- [19] WONG, G. T. Flight simulation & testing (performance). GDP individual report, 2020.
- [20] YIN, K. L. Empennage and rotor blade tip design of synchropter concept for mid-air retrieval mission. GDP individual report, 2020.

A The BEM Code

```

1  % Blade Element Momentum (BEM) Code
2  % (C) Enson Un, Jun 2020
3
4  clear
5  clc
6  close all
7
8  % Control inputs
9  theta0 = deg2rad(17.5);
10 thetas = deg2rad(-6.45);
11 thetac = deg2rad(1.1);
12
13 % Rotor parameters
14 N = 2; %number of blades
15 R = 4.023; %radius [m]
16 root = 0.15*R; %root radius [m]
17 c = @(r) 0.5747 + 0*r/R; %chord [m]
18 theta = @(r,psi) theta0 + thetas*sin(psi') + thetac*cos(psi') -deg2rad(11)*r/R...
    -deg2rad(0.08*6)*(r/R-0.92).*(r≥0.92*R); %pitch angle [rad]
19 EI = 5e5; %initial guess!
20 dm = 0.3; %initial guess!
21
22 % Aerofoil data
23 [cl,cd] = aerofoilData();
24
25 % Simulation settings
26 rho = 0.5328*1.225; %atm density [kg/m^3]
27 sos = 316; %speed of sound [m/s] (T=320K @5000m)
28 Vx = 62.5; %Vx from front; Vz downward [m/s]
29 Vz = 0.3;
30 omega = 51.2168; %rotation speed (=2*pi/T) [rad/s]
31 tipMach = omega*R/sos;
32
33 ns = 100; %spanwise discretisation
34 nt = 360; %wake discretisation
35 vi_relax_factor = 0.5;
36 maxharmon = 31; %size(maxharmon) > size(nwcoef) (must be odd, at least 3)
37 nwcoef = 6; %only the highest nwcoef of Fourier coefficients will be taken
38
39 % END OF INPUTS =====
40
41 r = linspace(root,R,ns);
42 psi = linspace(0,2*pi,nt);
43 localc = c(r);
44 localtheta = theta(r,psi);
45 dr = mean(diff(r));
46
47 wpsi = [ones(nt,1),cos((1:(maxharmon-1)/2).*psi'),sin((1:(maxharmon-1)/2).*psi...
    ')];
```

```

48 dwpsidpsi = [zeros(nt,1), (1:(maxharmon-1)/2).*sin((1:(maxharmon-1)/2).*psi')...
    ,-(1:(maxharmon-1)/2).*cos((1:(maxharmon-1)/2).*psi')];
49 d2wpsidpsi2 = [zeros(nt,1), -(1:(maxharmon-1)/2).^2.*cos((1:(maxharmon-1)/2).*...
    psi'), -(1:(maxharmon-1)/2).^2.*sin((1:(maxharmon-1)/2).*psi')];
50
51 % Rotor solidity
52 bladeArea = N*0.5*sum( (localc(1:end-1)+localc(2:end)).*(r(2:end)-r(1:end-1)) ...
    );
53 solidity = bladeArea/(pi*R^2);
54 localSol = N*localc/(pi*R);
55
56 % Initial guesses
57 mu = Vx/omega/R;
58 lambda = Vz/omega/R;
59 vi = ones(nt,ns); %induced velocity
60 wr0 = zeros(1,ns); %flapping displacement @ psi = 0
61 dwdr0 = zeros(1,ns); %bending gradient @ psi = 0
62 wcoef = zeros(1,maxharmon);
63
64 SF = zeros(nt,ns);
65 BM = zeros(nt,ns);
66 dwdr = zeros(nt,ns);
67 w = zeros(nt,ns);
68 newwcoef = zeros(1,maxharmon);
69
70 % CODE BODY =====
71
72 iter = 0;
73 while iter < 50
74     % Angles
75     vflap = omega*wr0.*sum( wcoef.*dwpsidpsi ,2 );
76     beta = atan(dwdr0.*sum( wcoef.*wpsi ,2 ));
77     phi = atan2(Vz + vi + vflap, Vx*sin(psi') + omega*r); %inflow angle
78     alpha = localtheta - phi;
79     sweep = atan2(Vx*cos(psi') , Vx*sin(psi') + omega*r);
80
81     % Forces
82     vSqr = (Vz + vi + vflap).^2 + (Vx*sin(psi') + omega*r).^2 + (Vx*cos(psi'))...
        .^2;
83     localM = sqrt(vSqr)/sos;
84     % Prandtls tip loss model
85     F = 2/pi*acos(exp(N*(r/R-1)./(2*lambda+2*mean(vi,'all')/omega/R)));
86     % Local forces at aerofoil section
87     Fz = 0.5*rho*F.*localc.*vSqr.*( cl(alpha,localM.*cos(sweep)).*cos(sweep).*...
        cos(phi) - cd(alpha,localM.*cos(sweep)).*sin(phi));
88     Fx = 0.5*rho*localc.*vSqr.*( cl(alpha,localM.*cos(sweep)).*cos(sweep).*sin...
        (phi) + cd(alpha,localM.*cos(sweep)).*cos(phi));
89     Fr = -sin(beta).*Fz;
90     % Global forces
91     dT = N*Fz.*cos(beta);
92     dH = N*(Fr.*cos(psi') + Fx.*sin(psi'));
93     dY = N*(Fr.*sin(psi') - Fx.*cos(psi'));
94     dP = N*omega*r.*Fx;

```



```

95     %dT(r > R*B) = 0;
96     T = 0.5*sum( (dT(:,1:end-1)+dT(:,2:end))*dr,2 );
97     H = 0.5*sum( (dH(:,1:end-1)+dH(:,2:end))*dr,2 );
98     Y = 0.5*sum( (dY(:,1:end-1)+dY(:,2:end))*dr,2 );
99     P = 0.5*sum( (dP(:,1:end-1)+dP(:,2:end))*dr,2 );
100    %B = 1 - 2*log(2)*sqrt(T/(rho*omega^2*pi*R^4))/N;
101    meanT = mean(T);
102
103    % Global moments
104    dMy = -N*r.*Fz.*cos(psi');
105    dMx = N*r.*Fz.*sin(psi');
106    My = 0.5*sum( (dMy(:,1:end-1)+dMy(:,2:end))*dr,2 );
107    Mx = 0.5*sum( (dMx(:,1:end-1)+dMx(:,2:end))*dr,2 );
108
109    % Vi =====
110
111    exitToken = 0;
112    % Update vi (Momentum theory)
113    meanvi = meanT/(2*rho*pi*R^2)/sqrt(Vx^2+(Vz+mean(vi,'all'))^2);
114    if abs(mean(vi,'all') - meanvi) < 0.005
115        exitToken = exitToken + 1;
116    else
117        vi = vi_relax_factor*vi + (1-vi_relax_factor)*meanvi*(ones(nt,ns)...
118            +tan(atan(mu/(lambda+meanvi/omega/R))/2).*r/R.*cos(psi') -2*mu*r/...
119            R.*sin(psi') );
120    end
121
122    % W =====
123
124    % Update beta: EI*d^4w/dr^4 + dm*d^2w/dt^2 = Fz
125    d2wdpsi2 = wr0.*sum( wcoef.*d2wpsidpsi2, 2 );
126
127    % Radial integration
128    for j = ns-1:-1:1
129        SF(:,j) = SF(:,j+1) + 0.5*(Fz(:,j+1)+Fz(:,j)) ...
130            - dm.*omega^2*(d2wdpsi2(:,j+1)+d2wdpsi2(:,j))*dr;
131        BM(:,j) = BM(:,j+1) + 0.5*(SF(:,j+1)+SF(:,j))*dr;
132    end
133    for j = 2:ns
134        dwdr(:,j) = dwdr(:,j-1) + 0.5*(BM(:,j-1)+BM(:,j))./EI*dr;
135        w(:,j) = w(:,j-1) + 0.5*(dwdr(:,j-1)+dwdr(:,j))*dr;
136    end
137
138    % Find Fourier coef
139    newwcoef(1) = mean(w(:,end));
140    for i = 2:(maxharmon-1)/2+1
141        newwcoef(i) = 2*mean(w(:,end).*cos((i-1)*psi'));
142        newwcoef(i+(maxharmon-1)/2) = 2*mean(w(:,end).*sin((i-1)*psi'));
143    end
144    if w(1,end) ≠ 0
145        newwcoef = newwcoef/w(1,end);
146    else
147        newwcoef = newwcoef/w(floor(nt/2),end);

```

```

147     end
148
149     [J,I] = sort(abs(newwcoef));
150     newwcoef(I(1:end-nwcoef)) = 0;
151
152     if any(abs(newwcoef - wcoef)) > 0.001
153         wcoef = newwcoef;
154     else
155         wcoef = newwcoef;
156         break
157     end
158
159     if w(1,end) ≠ 0
160         if all( abs(wr0-w(1,:)) < 1e-4 )
161             exitToken = exitToken + 1;
162         else
163             wr0 = w(1,:);
164             dwdr0 = dwdr(1,:);
165         end
166     else
167         if all( abs(wr0-w(floor(nt/2),:)) < 1e-4 )
168             exitToken = exitToken + 1;
169         else
170             wr0 = w(floor(nt/2),:);
171             dwdr0 = dwdr(floor(nt/2),:);
172         end
173     end
174
175     % EXIT CRITERIA =====
176
177     % Check exit criteria
178     iter = iter + 1;
179     if exitToken == 2
180         break;
181     end
182 end
183
184 meanH = mean(H);
185 meanY = mean(Y);
186 meanP = 1.39*mean(P);
187 meanMy = mean(My);
188 meanMx = mean(Mx);
189 eta = meanT*(Vz+mean(vi,'all'))/meanP;

```

B Power Lapse Curves

The engine power data are provided by Omar Arangath [1]. The graph is attached here for readers to check and compare against Fig.7, 8 and 9.

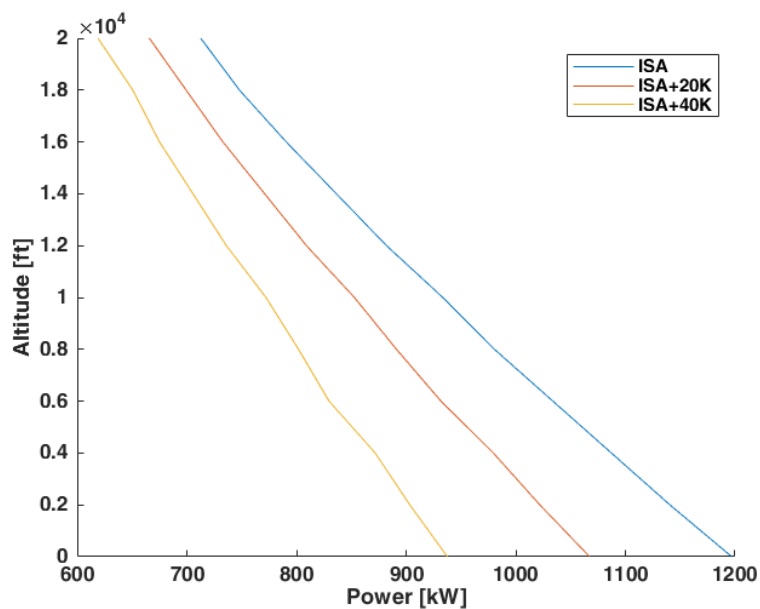


Figure 11: Power lapse curve at $\Delta T = 0K, 20K, 40K$